A REDUCED, QUASI-SEPARATED STACK WITH IRREDUCIBLE SPACE IS INTEGRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that a quasi-separated algebraic stack is decent. Invoked at the start of the counted proof (source lines 10694-10695). Supplies decency, one of the clauses in the definition of an integral stack.
- The base change permanence for quasi-compactness of the diagonal. Invoked inside the counted proof (source lines 10696-10699). Shows that the inertia stack is quasi-compact over the stack.
- The definition of an integral algebraic stack. Used at the end of the counted proof (source lines 10700-10703). Collects the hypotheses: reduced, decent and with quasi-compact inertia.

Source reference. Original definition definition-integral-algebraic-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10702–10702. Navigation pointer only: the counted text refers to the Stacks item definition-integral-algebraic-stack, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-base-change-quasi-compact: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2062–2062. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-quasi-compact, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-quasi-separated-decent: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 10695–10695. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-separated-decent, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack which is reduced and quasi-separated and whose associated topological space $|\mathcal{X}|$ is irreducible. Then $\mathcal{X}$ is integral.*

Proof. If $\mathcal{X}$ is quasi-separated, then $\mathcal{X}$ is decent by Lemma lemma-quasi-separated-decent. If $\mathcal{X}$ is quasi-separated, then $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is quasi-compact, hence $\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is quasi-compact as the base change of Δ by Δ, see Lemma lemma-base-change-quasi-compact. Thus we see that all the hypotheses of Definition definition-integral-algebraic-stack hold (and we also see that we may replace "quasi-separated" by "$\Delta_\mathcal{X}$ is quasi-compact"). □